

Key Points:

- We estimate homogenous lock-in functions for several lacustrine and marine sediment records
- We propose a new method to transform relative declinations to absolute values
- The large variety of estimated lock-in functions indicates the importance of taking post-depositional detrital remanent magnetization distortions into account

Supporting Information:

Supporting Information may be found in the online version of this article.

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Estimating Post-Depositional Detrital Remanent Magnetization (pDRM) Effects for Several Lacustrine and Marine Sediment Records Using a Flexible Lock-In Function Approach

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Abstract Geomagnetic field models over past millennia rely on two main data sources: archeomagnetic data provide snapshots of the geomagnetic field at specific locations, and sediment records deliver time series of the geomagnetic field at specific locations. The limited temporal and spatial coverage of archeomagnetic data necessitates the incorporation of sediment data especially when models go further back in time. When working with sediment data one should consider the post-depositional detrital remanent magnetization (pDRM) process, which can cause delayed and smoothed signals. To address the distortion associated with the pDRM process a Bayesian modeling technique incorporating archeomagnetic data and a class of flexible parameterized lock-in functions has been proposed. In this study, we investigate this method in more detail and apply it to declination and inclination of several lacustrine and marine sediment records. Data-driven results support the hypothesis that the pDRM process can introduce distortions, including offsets and smoothing, in some lacustrine and marine sediment records. We demonstrate a correction approach to minimize the distortion caused by the pDRM process and its impact on geomagnetic field reconstructions. The variability in the results observed across the nine records points to a potential dependence on sedimentological characteristics. To explore this further, we plan to systematically apply our novel method to a larger number of records in future studies.

Plain Language Summary Understanding the Earth's magnetic field changes over the past millennia helps us learn more about the planet's history. We can use the magnetic field information preserved in different materials to reconstruct the past Earth's magnetic field evolution: ancient artifacts (archeomagnetic data), lava flows and sediment records. Archeomagnetic and lava flow data has limited coverage in time and space, so when modeling longer time scales we have to rely more and more on sediment data. When using sediment data, we need to be careful about a process called post-depositional detrital remanent magnetization (pDRM), which can distort the signals and cause smoothing and delays. To deal with this issue, a new method has been developed using Bayesian modeling, archeomagnetic data and flexible lock-in functions to correct for the pDRM effect. In this study we explore this method in more detail and test it on several lacustrine and marine sediment records. We found that the pDRM process indeed introduces distortions in several sediment records.

1. Introduction

Understanding the geomagnetic field over the last millennia depends on the integration of two primary data sources. The first encompasses archeomagnetic samples and volcanic lava flow samples, which provide discrete snapshots of the geomagnetic field at specific locations and times. These materials acquire their magnetization through the process of thermoremanent magnetization (TRM), capturing a clear and instantaneous signal of the geomagnetic field as they cool from a state above the Curie temperature (e.g., Stacey, 2012). The second class consists of sediment records, offering time series of geomagnetic field changes at particular sites over extended periods. Magnetization in sediments, first measured by McNish and Johnson (1938), is acquired through detrital remanent magnetization (DRM) processes, where sedimentary particles align with the geomagnetic field during and after deposition and consolidation. These data sets have been used to derive various geomagnetic field models, each with varying degrees of resolution and temporal scope (e.g., Arneitz et al., 2019; Constable et al., 2016; Hellio & Gillet, 2018; Nilsson & Suttie, 2021; Schanner et al., 2022).

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While both classes of geomagnetic data are important, they are not without their challenges. Archeomagnetic data suffer from a significant limitation in their distribution across time and space. Temporally, the further we go back into the past, the more sparse the data becomes, making older periods less represented. Spatially, the Northern Hemisphere is much better covered, with big clusters in Europa, Central and North America and East Asia, while we face limited number of data in the Southern Hemisphere. This uneven distribution leads to models that may be geographically biased or incomplete.

Sediment records, on the other hand, offer a better temporal and spatial coverage, but pose different challenges. The process by which sediments acquire their magnetization, is influenced by a variety of environmental and post-depositional factors, such as the interaction of magnetic particles with the substrate at the sediment-water interface and the subsequent dewatering of the sediment (Irving, 1957). Moreover, technical aspects like the loss of orientation during core drilling lead to relative values instead of absolute ones.

The exact mechanisms by which sediments acquire their remanent magnetization are still poorly understood, and the terminology describing these processes has often been inconsistent across different studies. In our work, we adhere to the classification outlined by Verosub (1977), which distinguishes between depositional DRM (dDRM) and post-depositional DRM (pDRM). dDRM encompasses errors such as the inclination error, a result of the preferential alignment of non-spherical particles as they settle (R. King, 1955; Griffiths et al., 1960).

Early models of pDRM suggest that, immediately after deposition, coarse-grained particles are rapidly immobilized, while finer grains remain in flux within the sediment matrix. Over time and under progressive sedimentary consolidation, these finer particles also experience a mechanical lock-in (e.g., Hamano, 1980; Irving, 1957; Irving & Major, 1964; Kent, 1973; Otofujii & Sasajima, 1981). However, alternative theories challenge this model, suggesting that sediment cohesion limits grain mobility (Katari et al., 2000) and that pDRM acquisition might be driven by bioturbation (e.g., Egli & Zhao, 2015).

Despite the lack of consensus on the finer details of these processes, what is broadly agreed upon is the resulting impact on the geomagnetic signal—a delayed and smoothed representation of the magnetic field. The recording process can be represented by the convolution of the original magnetic field variations with a kernel, called lock-in function, over the time required for the remanence carriers to reach their definitive orientation (Roberts & Winklhofer, 2004; Suganuma et al., 2011).

In Bohsung et al. (2023) a new class of flexible lock-in functions capable of modeling the delay and smoothing related to the pDRM process was presented. Depending on four parameters these lock-in functions can approximate a wide range of possible lock-in behaviors. To determine these parameters for a given sediment record a Bayesian modeling technique based on Gaussian Processes was employed. The essence of this method is to utilize independent magnetic information from archeomagnetic data and lava flows. Together with prior assumptions on the geomagnetic field this information is used in a Bayesian inference technique to estimate the lock-in function parameters for a single sediment record. Internally, a global process based on the archeomagnetic data and the prior assumptions is used as a reference for the sediment record. While estimating the lock-in function parameters, the method respects prior uncertainties, measurement uncertainties and temporal uncertainties for archeomagnetic data and the sediment record. The method does not rely on nor produce a (local) geomagnetic field model; instead, it directly harnesses the raw, independent measurements from archeomagnetic data and lava flow samples.

In this paper we apply the proposed method to real-world lacustrine and marine sediment records from various globally distributed locations. We focus on sediment records covering Holocene time periods. The utilization of real sediment data necessitates the comprehensive consideration of factors extending beyond the inherent distortions attributed to the pDRM process. Rigorous and careful data selection and preprocessing procedures are crucial. This includes aspects such as estimation of declination offsets, considering inclination shallowing effects, the formulation of a robust age-depth model and the judicious identification and exclusion of outliers. Within the scope of this study, our primary focus remains centered on the comprehensive examination of the pDRM process effects and the newly proposed method. We handle calibration of declination offsets as well as inclination shallowing as a part of the preprocessing procedure.

In Section 2, we summarize the method proposed in Bohsung et al. (2023), and we introduce a novel approach to calibrate relative declinations. A list of nine sediment records is presented in Section 2.5. The preprocessing of

these records includes a new method for estimating the offset required to transform relative declinations into absolute declination values, as well as the construction of updated age-depth models, using the most recent radiocarbon calibration curves. In Section 3, we apply the method to these sediment records and present the results, which then are discussed and interpreted in Section 4.

2. Method and Materials

This section is structured as follows. We begin this section with a simplified overview of the method from Bohsung et al. (2023), giving the reader a broad understanding. Subsequently, we will dive into specifics, providing details about the parameter estimation. In Section 2.2, we introduce a novel approach for estimating declination offsets, which is crucial to transform relative into absolute declinations. Lastly, we discuss the creation of age-depth models for our sediment data and other preparation/preprocessing steps required for our analysis.

We start by summarizing the findings of Bohsung et al. (2023). Our aim here is to provide a general picture of the method, focusing on its objective to identify parameters that effectively characterize the distortions associated to pDRM effects. It is essential to note that our approach does not preprocess the raw sediment data nor does it propose a new geomagnetic field model. Instead, it seeks to identify parameters that represent the pDRM effects, namely the offset and smoothing seen in sediment data.

The smoothed and delayed signal in the sediment data can be described by a convolution or weighted average of the geomagnetic field signal, $\mathbf{B}$, and a lock-in function, F , over the lock-in period λ

$$\mathbf{M}(z) = \int_0^\lambda \mathbf{B}(z - z') F(z') dz' \quad (1)$$

where $\mathbf{M}(z)$ represents the resultant magnetization of the layer at depth z (Roberts & Winkhofer, 2004; Suganuma et al., 2011). The weights are given by a lock-in function F . To understand the pDRM effects on the magnetic signal in the sediment record we have to investigate F . An important assumption here is that the lock-in function is homogenous over a given period. This assumption's validity diminishes with the extension of the time period and the presence of substantial lithostratigraphical variations in the sediment record.

To tackle the complexity of pDRM effects Bohsung et al. (2023) introduced a new class of flexible lock-in functions capable of approximating a wide range of possible lock-in behaviors and previously suggested lock-in functions. The class of piecewise linear parameterized lock-in functions is given as

$$F_{b_1, b_2, b_3, b_4}(z) = \frac{2}{-b_1 - b_2 + b_3 + b_4} \begin{cases} 0 & z \leq b_1 \\ \frac{z - b_1}{b_2 - b_1} & b_1 < z \leq b_2 \\ 1 & b_2 < z \leq b_3 \\ \frac{b_4 - z}{b_4 - b_3} & b_3 < z \leq b_4 \\ 0 & b_4 \leq z \end{cases} \quad (2)$$

Depending on the four non-negative and sequentially ordered parameters b_1, b_2, b_3, b_4 , the parameterized function F_{b_1, b_2, b_3, b_4} can model the offset as well as the smoothing associated to the lock-in process. By approximating the lock-in behavior by a lock-in function from this class we can estimate the effects associated to pDRM by estimating the four parameters and therefore the lock-in function for a given sediment record.

To estimate the four parameters we use a Bayesian modeling technique based on Gaussian processes. As visualized in Figure 1, the method is initialized by the following inputs: (a) independent magnetic information in form of a global compilation of archeomagnetic data and lava flow samples; (b) prior assumptions about the geomagnetic field; (c) the prepared/preprocessed data from a single sediment record, that is, with calibrated declination, depth-to-age calculations etc (see Section 2.4 for more details). Additionally, a set of parameters, $b_1, \dots, b_4$, is provided (red). The associated lock-in function is then used internally to convolve the signal (blue)

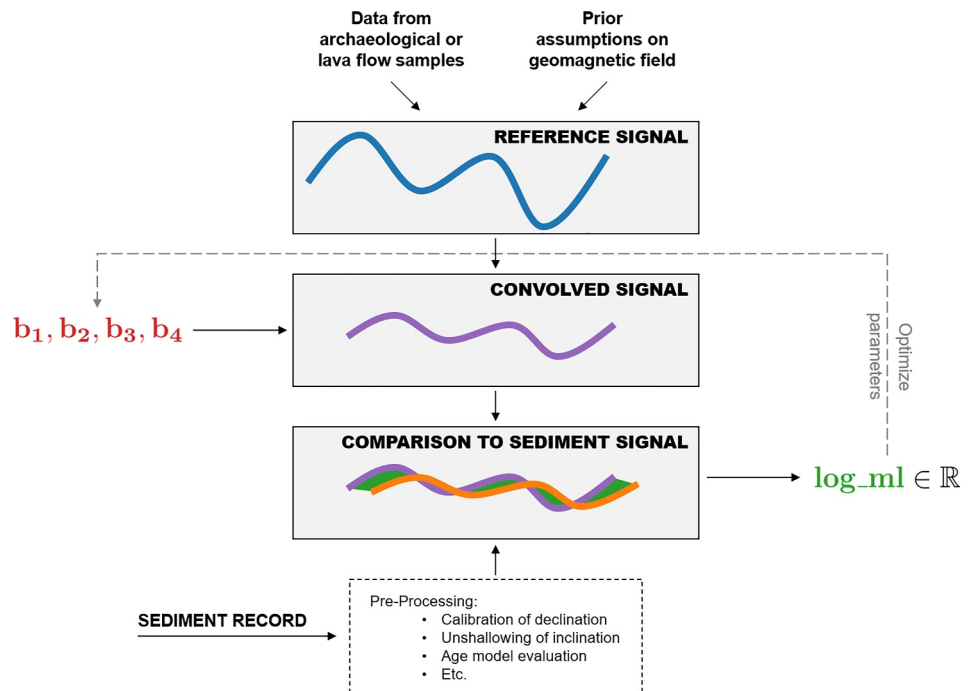


Figure 1. Simplified visualization of the proposed method. Internally, the method convolves the combined signal of independent magnetic information and prior assumptions (blue) with a set of lock-in function parameters (red). The convolved signal (purple) is then compared to the signal for a single sediment record (orange). The resulting $\log_ml$ value (green) can be interpreted as a measure of agreement for a given set of parameters, $b_1, \dots, b_4$. By optimizing the $\log_ml$ value we can find the best set of parameters.

derived from the independent magnetic information and the prior assumptions using the formula in (1). The convolved signal (purple $\sim$ blue and red) is compared to the sediment data signal (orange). The method returns a single value which can be seen as a measure of agreement between the two signals (green). In other words, for given archeomagnetic and sediment data sets and prior assumptions we have a function that takes four parameters $b_1, \dots, b_4$ as inputs and returns a single value. In the framework of Bayesian statistics this function is called log-marginal likelihood (log ml). By optimizing this function we identify the set of lock-in function parameters that best accounts for pDRM effects. It is important to note that this very simplified description of the workflow is streamlined for ease of understanding and not for formal correctness.

2.1. Estimation of the Lock-In Function Parameters

In this section, we dive into more details of the summary of parameter estimation. Theoretical formulations and more details can be found in Bohsung et al. (2023) and corresponding references given below.

To estimate the parameters $b_1, \dots, b_4$, we use the Kalman filter (Kalman, 1960) based method described in Bohsung et al. (2023). In this section we will shortly summarize how the method works. For the estimation we perform a type-II maximum likelihood estimation (Rasmussen, 2004). While closed-form marginal likelihood is available for Gaussian processes (Rasmussen, 2004), its numerical evaluation becomes impractical given the extensive archeomagnetic data set used. Therefore, we adopt a sequentialized marginal likelihood evaluation inspired by previous works (Baerenzung et al., 2020; Särkkä, 2013; Schanner et al., 2022). The marginal likelihood is approximated as a sum over values calculated for individual Kalman filter steps, providing a measure of how well a set of lock-in function parameters describes the post-depositional remanent magnetization (pDRM) process in a sediment record. This estimation leverages global archeomagnetic and volcanic data as well as data from a single sediment record, focusing on the last eight thousand years due to the temporal distribution of the archeological data set.

A notable difference from existing implementations is the incorporation of a modified observation functional (see Bohsung et al. (2023) for details). This modification accommodates cross correlations between the Kalman filter steps resulting from the convolution integral that leads to a delay and smoothing in the measurements.

Choosing a spherical harmonics cutoff degree of $l_{\max} = 8$ and a Kalman filter step size of $\Delta t = 40$ years strikes a balance between estimation accuracy and computational efficiency. Tests demonstrate that variations in time steps and cutoff degrees minimally impact estimation accuracy while significantly increasing computational time.

To optimize the log-marginal likelihood (log-ml) we use the methodology including the 50 optimization runs outlined in Bohsung et al. (2023), utilizing dlib's LIPO-TR function optimization algorithm (Malherbe & Vayatis, 2017; D. E. King, 2009). However, in contrast to the synthetic tests performed in Bohsung et al. (2023), determining an appropriate upper bound for the parameter estimation for real sediment records is not straightforward a priori. To address this, we initiated the estimation process with an upper bound of 100 cm and performed 10 estimations. In cases where the maximal lock-in depths (values of parameter b_4) of these 10 estimated lock-in functions are significantly lower than 100 cm (<90 cm), we performed another 40 estimations. Conversely, if the estimated lock-in depths are too close to the upper bound we increased the upper bound by 100 cm. This incremental approach is advised to avoid potential issues where the optimizer converges to local optima.

In our parameter estimation process, we are optimizing the log-ml value. Implicitly we derive a global field model based on archeomagnetic data and data from a single sediment record. For a complete inversion one would need to store intermediate results and ideally apply a smoothing algorithm in order to incorporate full correlations between the individual steps. However, during the parameter estimation process we do not store and develop this model completely as our focus is on finding lock-in function parameters that maximize the log-ml value. In the synthetic tests performed in Bohsung et al. (2023), we assessed the accuracy of our estimated lock-in function by comparing it with the lock-in function used to generate the synthetic data. However, for real data, an alternative approach is required. To evaluate the effectiveness of the best-estimated lock-in function, we employ these parameters in a full inversion, where intermediate results are stored, and a smoothing algorithm is applied. Subsequently, the smoothing functional is applied to the posterior and the predictions of the smoothed posterior are compared with the sediment observation. This smoothed posterior, referred to as predicted sediment observations, serves as a benchmark, and the closer its alignment with the sediment observations, the more accurate the estimated lock-in function.

2.2. Estimation of the Declination Offset

In general, declinations of a sediment record are reported as relative values, primary due to the absence of core orientation during drilling. Different methods have been proposed to estimate the resulting offset (e.g., Nilsson et al., 2014; Panovska et al., 2015). In this paper, we present a new method based on Bayesian statistics, utilizing the global geomagnetic field model ArchKalmag14k (Schanner et al., 2022) as a reference model. In the future, we intend to incorporate the offset parameter as an extra hyperparameter estimated simultaneously with $b_1, \dots, b_4$. However, for the current investigation, our primary focus lies in analyzing the lock-in function. Consequently, we have made the decision to estimate the offset parameter as part of the preprocessing phase.

In contrast to the estimation of the lock-in function parameters, we use a pretrained model, that is, we use the posterior distribution of the ArchKalmag14k model and a Type-II maximum likelihood estimation to estimate the optimal offset. To incorporate prior information about the offset, we assume a univariate normal prior distribution with mean μ_O and variance σ_O^2 . Under the assumption that the mean field is an axial dipole, we set μ_O to the mean distance of the observed relative declinations to zero. This helps to guide the estimation process. In order not to constrain the optimization algorithm too strongly, we set $\sigma_O^2 = 180$.

2.3. Age-Depth Model

Many of the age-depth models published together with the original data do not report uncertainties. Some use paleo secular variations to complement radiocarbon dates. Further, most of them are constructed using out-dated radiocarbon calibration curves that is, older versions of IntCal for Northern Hemisphere, SHCal for Southern Hemisphere and/or Marine for oceanic records; see Reimer et al. (2020), Hogg et al. (2020) and Heaton et al. (2020) for the most recent curves and further details. To incorporate uncertainties from the age-depth

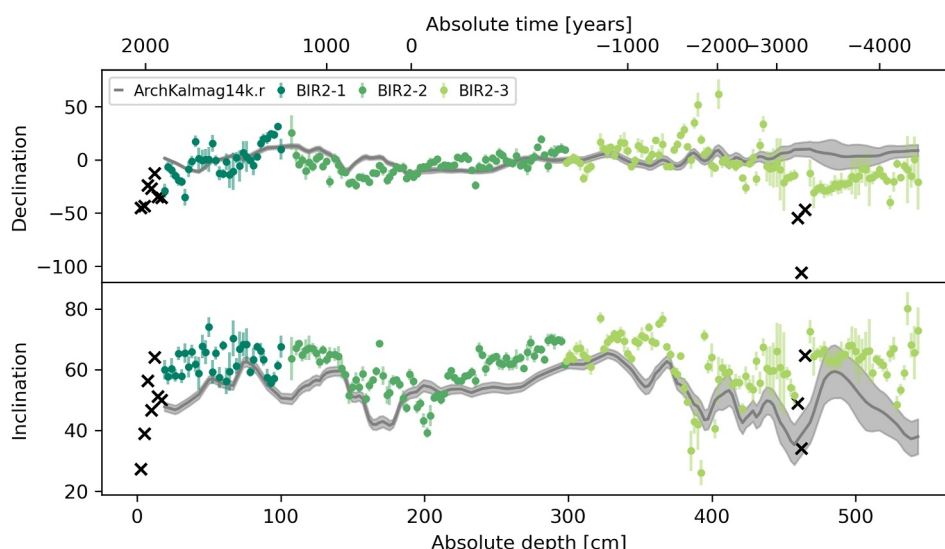


Figure 2. The figure shows declination and inclination of BIR color-coded by sub-sections together with predictions of ArchKalmag14k.r. Removed data points are shown as black crosses.

determination into our modeling and to prevent circular reasoning introduced by paleomagnetically dated sediment cores, we recalibrate all records based on available radiocarbon ages. We only use radiocarbon age information and ignore paleo secular tie points from the original studies to prevent circular reasoning. To model the age-depth relation, we pursue a Markov chain Monte-Carlo (MCMC) based approach. It is similar to the algorithm proposed by Nilsson and Suttie (2021), which itself is a modification of the *bacon* algorithm (Blaauw & Christen, 2011). Briefly, a piecewise linear model with correlated slopes is fit to the radiocarbon dates.

The depositional ages depend on additional lithological factors and may be influenced by bioturbation (Egli & Zhao, 2015), the so-called Brazil nut effect (Savranskaia et al., 2022) and other processes like differential dissolution (Barker et al., 2007). Including these effects in the age-depth modeling procedure is difficult due to many unknowns related to the processes. It is beyond the scope of this study to address this issue. We want to note here again, that imprecise dating related to the mentioned processes influences the quality of our findings. Effects from incorrect age-depth modeling may leak into the lock-in function as additional shifts or smoothing. We plan to model lock-in and age-depth relation jointly in the future, but this is a complex challenge beyond the scope of this study.

Due to a shift in the Marine20 curve (with respect to the older Marine13 curve), some age-depth models deviate by about one hundred years from the originally published ones. However, many original curves already contained an estimation of the local reservoir effect and therefore the shift in Marine20 does not translate to all age-depth models directly. The main benefit of recalibrating the age-depth models is the availability of dating uncertainties for all sediment records included in this study and the prevention of circular reasoning.

2.4. Data Preprocessing

The data preprocessing for each sediment record was performed as follows. Initially, the declination and inclination values for each record were plotted for visual outlier detection. Certain data points deviate strongly with respect to the others. In some cases not only single points but whole segments show significant increases or decreases in declination or inclination at the beginning or end of a core or sub-section. An example of such a case can be found in Figure 2 (black crosses). As the reasons behind these phenomena remain unexplained, such segments were removed from both inclination and declination. The black crosses at the beginning of the sub-core BIR2-1 show an unusual decrease. They were excluded from further analysis. Also, three data points from BIR2-3 are excluded as outliers in the declination.

Subsequently, the maximum angular deviation (MAD) values were converted to α_{95} values using the method outlined in Khokhlov and Hulot (2016). In cases where the number of demagnetization steps for certain sediment

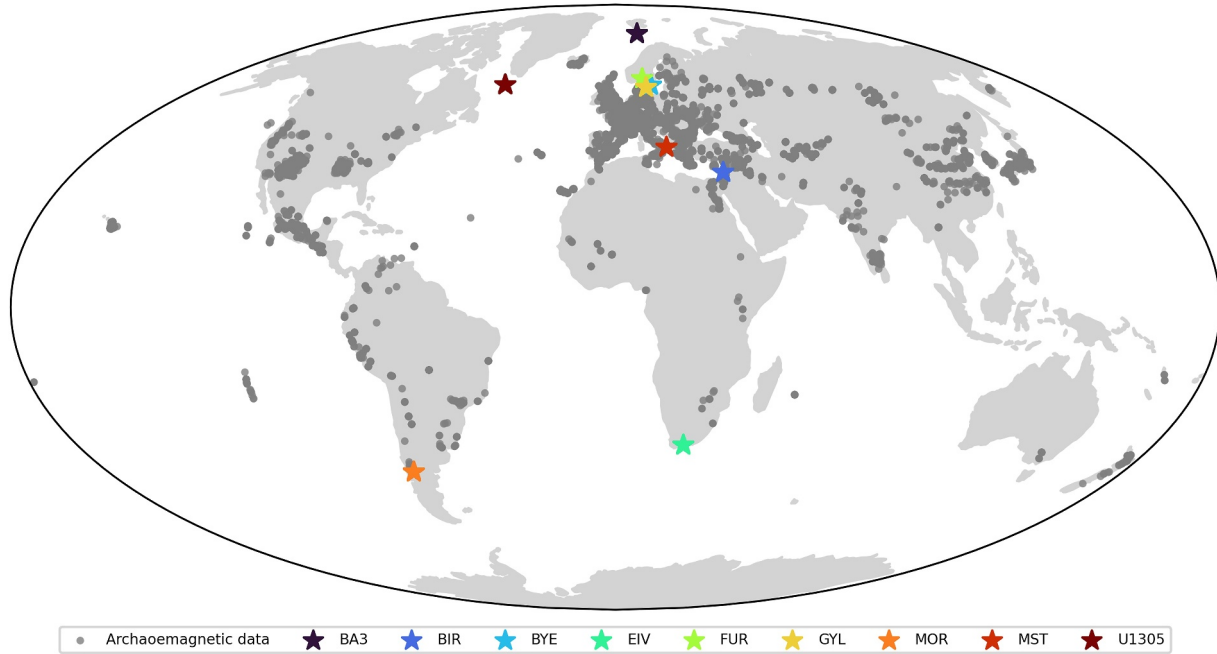


Figure 3. Spatial distribution of used data. Dark gray dots represent the archeomagnetic data and the sediment record locations are represented as colored stars.

records was unavailable in the associated publications, a default value of four steps was used. For data points with missing MAD values, a default α_{95} value of 7° was assigned. These α_{95} values were then employed to calculate the measurement errors in declination (D) and inclination (I) according to the formulas (Lanos et al., 2005; Suttie & Nilsson, 2019)

$$\alpha_I = \frac{57.3^\circ}{140} \alpha_{95} \quad \alpha_D = \frac{1}{\cos(I)} \alpha_I \quad (3)$$

For data points where only declination information was available, but inclination data was missing, an approximation method based on the dominance of the geomagnetic field's dipole nature was utilized. The inclination (I) can then be estimated using the formula

$$I \approx \tan^{-1}(2 \tan(\text{lat})) \quad (4)$$

where “lat” represents the latitude at the location of the sediment record.

Lastly, the offset parameter, necessary for converting relative declinations into absolute values, was estimated using the methodology proposed in Section 2.2.

2.5. Data

We use the same archeomagnetic data as in Schanner et al. (2022). The spatial distribution is shown in Figure 3 (gray dots).

The proposed method is applied to nine lacustrine and marine sediment records (see Figure 3 and Table 1). We focus on records with a clearly identifiable signal (visually distinct from noise) that does not show variation faster than expected from established secular variation models. Records with too flat signals are excluded as well as records with unreasonably small uncertainties, that is, reported uncertainties are clearly smaller than the data scattering or are on average smaller than one degree. Additionally, we chose sediment records with radiocarbon dating to generate independent age-depth models. Lithology or sedimentation rate played no role in the selection of the records. In this study we focus on directional data (declination and inclination) only. The main reasons are potentially poorer data quality due to environmental influences on the intensity signal, and missing uncertainty

Table 1
Overview of Used Sediment Records

Core ID	Sub-cores	Lat (deg)	Long (deg)	N	Time coverage	References	Source
BA3	–	74.850	14.802	434	8585 BCE–1496 CE	Caricchi et al. (2018) Caricchi et al. (2020)	GFZ Data Services
BIR	–	33.231	35.769	215	4308 BCE–1847 CE	Schwab et al. (2004) Frank et al. (2002) Frank et al. (2003)	Geomagia ^a
BYE	ByaP2, ByaP3	57.383	15.343	284	9263 BCE–75 BCE	Snowball and Sandgren (2004)	Geomagia ^a
EIL	–	–33.995	22.640	99	938 CE–1649 CE	Wündsche et al. (2016a)	Pangaea ^b Wündsche et al. (2016b)
FUR	P2, P3	59.383	12.080	242	6822 BCE–1966 CE	L. Zillén et al. (2003) L. M. Zillén et al. (2002)	Geomagia ^a
GYL	GP1, GP2, GP4	56.759	13.177	419	4494 BCE–1178 CE	Mellström et al. (2013) Snowball et al. (2013)	Geomagia ^a
MOR	–	–41.000	288.5	147	10298 BCE–1329 CE	Gogorza et al. (2000)	Paper
MST	–	39.834	17.801	138	2009 BCE–477 BCE	Béguin et al. (2019)	Author
U1305	–	57.475	311.471	147	6049 BCE–1135 CE	Stoner et al. (2013a)	Stoner et al. (2013b)

^aBrown et al. (2015). ^bFelden et al. (2023).

information. See Bohsung et al. (2023) for a discussion of this point. In a future study we will investigate the lock-in behavior of intensities and compare it to the results for directional data. Detailed information about the individual sediment records and especially about their preprocessing is given in Section 3.

3. Results

In this section, we present the results obtained from estimating the lock-in function parameters. Each sediment record is analyzed individually, and the results are separately visualized in Figures 4–7 and 9 as well as in Figures S1–S11 in Supporting Information S1. These figures provide the following information.

The 50 estimated lock-in functions are visualized in (a). The lock-in function with the highest (best) log-ml value is highlighted in orange and was used for the inversion. The remaining estimated lock-in functions are color-coded, ranging from red (indicating a low log-ml value) to blue (indicating a high log-ml value). The distribution of the log-ml values is illustrated in (d). To assess the accuracy of our predictions, we present the predicted sediment observations (mean and one hundred samples) for declination and inclination in the upper and lower panels of (b), respectively. These predictions are derived from the posterior, generated using the lock-in function highlighted in orange (see Section 2.1 for details). Furthermore, the directional paleomagnetic records are shown along with their respective measurement errors. In cases where a sediment record consist of multiple sub-cores or sub-segments we distinguish them by different colors. Additionally, the posterior mean and uncertainties of ArchKalmag14k.r is visualized in gray. Note that we are not correcting the data. To correct the data one would need to apply the set of estimated lock-in function parameters in the deconvolution of 1, which will be done in a future study. To examine the characteristics of the lock-in functions, we employ density plots in (e) and (f), which demonstrate the distributions of two parameters: half lock-in depth $\rho_{0.5}$, and lock-in function width ρ_w , as described in Bohsung et al. (2023). The half lock-in depth $\rho_{0.5}$ denotes the depth where half of the lock-in process is completed and can be interpreted as the horizontal offset induced by the pDRM effect. Lock-in function width ρ_w can be associated to the smoothing induced by pDRM effects. In Bohsung et al. (2023) a third parameter, the lock-in function height was investigated. We decided to exclude this parameter from further consideration due to imprecise determination and no interpretation power. The density estimation of $\rho_{0.5}$ and ρ_w is conducted using Gaussian kernel density estimation. The parameter associated with the lock-in function that yields the best log-ml value is highlighted in orange. Finally, (c) shows the mean (red) and 50 samples (gray) of the age-depth model. Radiocarbon ages are depicted as violin plots. The color of the violins indicates which type of calibration curve

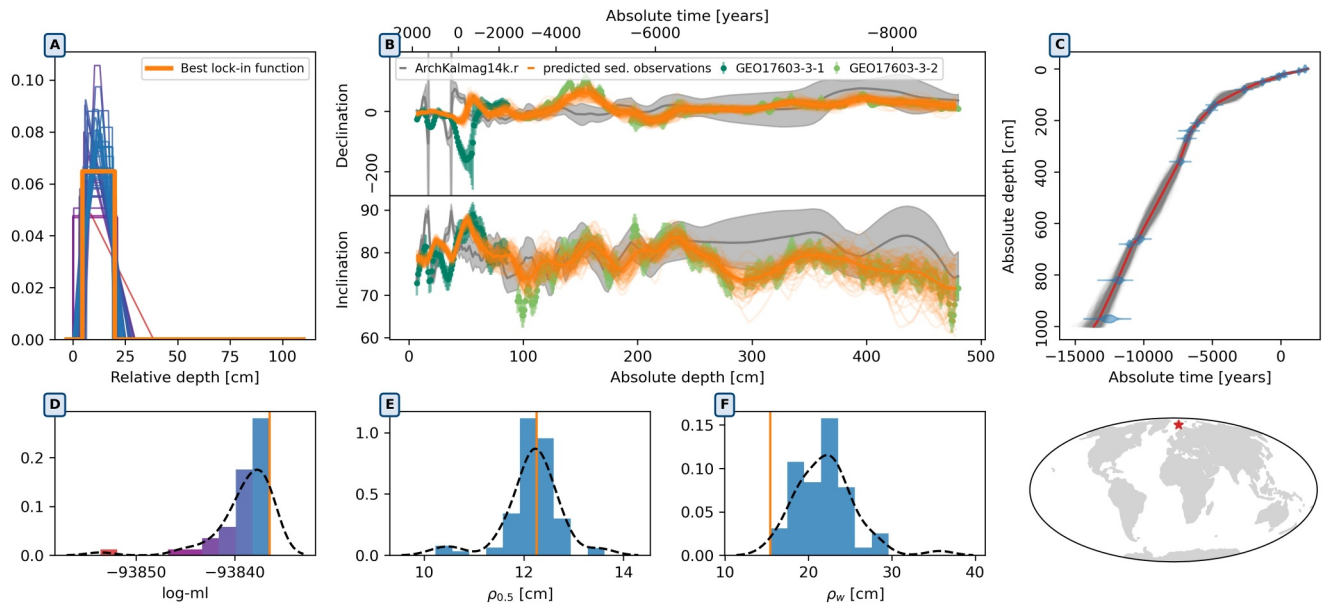


Figure 4. Results for BA3. (a) 50 estimated lock-in functions, with the lock-in function yielding the highest log-ml value highlighted in orange and used for inversion. The remaining functions are color-coded, ranging from red (low log-ml value) to blue (high log-ml value), as illustrated in (d). To assess prediction accuracy, we show the mean (thick orange line) and one hundred samples (thin orange lines) of predicted sediment observations for declination and inclination in upper and lower panel of (b), derived from the posterior using the orange lock-in function. The directional paleomagnetic records, along with their respective measurement errors, are depicted, in shades of green distinguishing sub-cores or sub-segments. The gray visualization shows the posterior mean (solid line) and uncertainties (gray area) of ArchKalmag14k.r. (e) and (f) Density plots of two lock-in function parameters: $\rho_{0.5}$, ρ_w , in centimeter. The parameter with the best log-ml value is highlighted in orange. (c) Mean (red) and 50 samples (gray) of the age-depth model, with radiocarbon ages depicted as violin plots, color-coded by the type of calibration curve used (blue for marine and orange for land curves). Global location of the sediment record is visualized in the bottom right corner.

was used for the individual ages, blue corresponding to the marine and orange to the respective land curve. In most cases the decision which type of curve to use for which radiocarbon sample was guided by the original publication, with bulk sediments being calibrated by the marine curve and plant or wood remnants by the land curves. The spatial location of the sediment record is visualized in the bottom right corner.

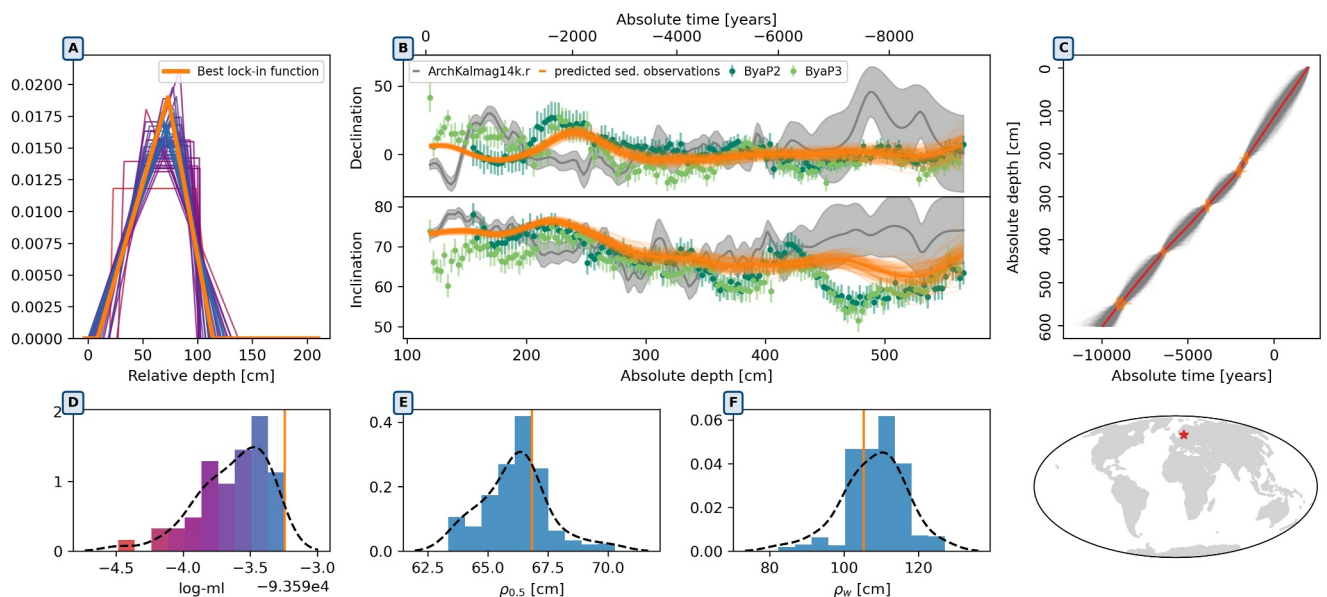


Figure 5. Results for BYE.

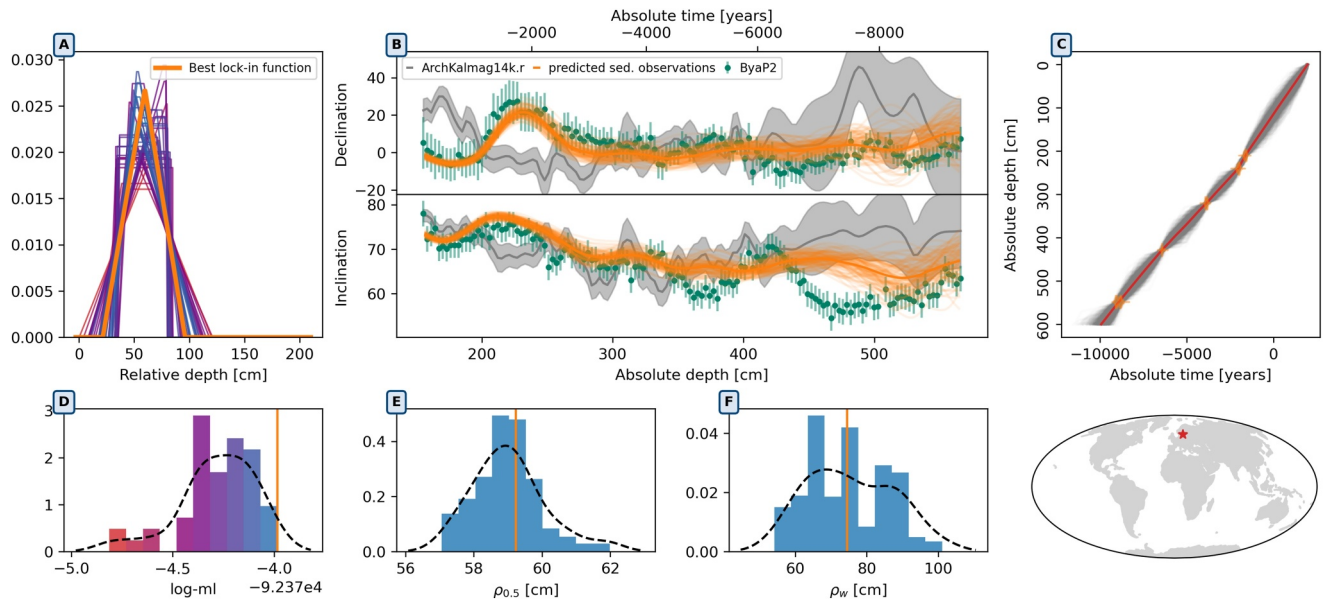


Figure 6. Results for ByaP2.

Sediment records BYE, FUR, and GYL consist of at least two different sub-cores. For each of them we applied the proposed method not only to the combined records but also to the individual sub-cores.

In the following, we will present the results for BA3 and BYE. For the results of the remaining records see Supporting Information S1.

3.1. BA3

Removed data points: Due to an abrupt change observed at a depth of 481 cm, as well as unusual variations in the declination, all data points below this depth were excluded from further analysis.

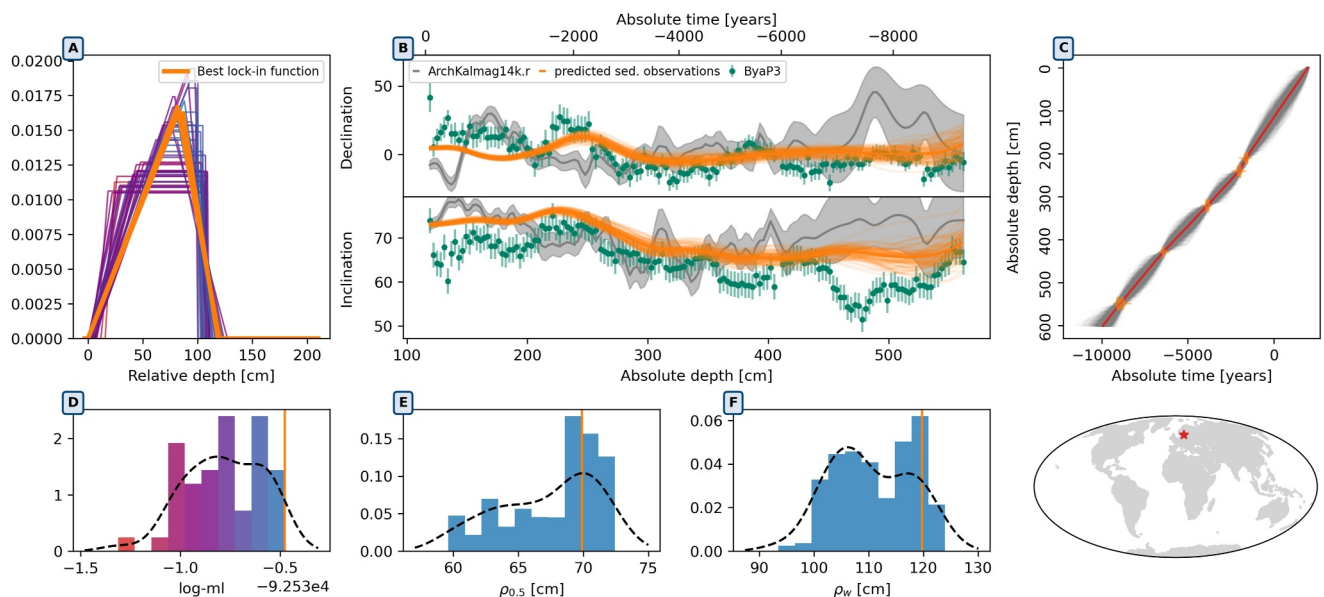


Figure 7. Results for ByaP3.

Declination offset: The sediment record was divided into two sub-sections, namely GEO17603-3-1, which covers the upper part of the record down to a depth of 83 cm, and GEO17603-3-2, which covers the lower part. The estimated declination offset values are $-2.46^\circ \pm 0.39$ for GEO17603-3-1 and $-26.04^\circ \pm 0.57$ for GEO17603-3-2.

Age-depth model: ^{14}C ages were taken from the original publication, Table 3 (Caricchi et al., 2018). All samples were calibrated using Marine20. See Figure 4 plot (C) for the resulting age-depth model.

Estimation: The estimation was performed using an upper bound of 100 cm. Parameter b_4 is significantly lower compared to this upper bound across all 50 estimated lock-in functions, indicating that the chosen bound of 100 cm was appropriate. Furthermore, the predicted sediment observations demonstrated a strong correspondence with the actual data, indicating a good fit. Results are visualized in Figure 4.

3.2. BYE

Removed data points: The first 14 data points from the upper part of the sub-core ByaP2 have been removed due to an abrupt decrease in declination.

Declination offset: For the sub-core ByaP2 we estimated an offset of $-1.18^\circ \pm 0.9$ and for ByaP3 an offset of $-0.08^\circ \pm 1.02$.

Age-depth model: ^{14}C ages were taken from the original publication, Table 1 (Snowball & Sandgren, 2004). All samples were calibrated using IntCal20.

Estimation combined record: First, we conducted 10 estimations using an upper bound of 100 cm. Since the parameter values b_4 were too close to this upper bound we decided to increase it to 200 cm. With this new upper bound all parameters b_4 were found to be significantly lower than 200 cm, indicating that this revised upper bound was appropriate. The predicted sediment observations demonstrated a strong correspondence with the declination. Except for the interval between 450 and 530 cm, the predicted sediment observations also aligned well with the inclination.

Estimation sub-cores: We can observe a strong agreement in the results obtained for the two sub-cores (ByaP2 and ByaP3, compare Figures 6 and 7) and the combined data (BYE, see Figure 5). Remarkably, the comparison of the three parameters $(\rho_{0.5}, \rho_h, \rho_w)$ derived from the best estimated lock-in functions reveals that the combined record's values fall within the range delineated by the corresponding values of the two sub-cores. These observations indicate a consistent signal in the individual sub-cores and the combined record.

For more examples see Supporting Information S1.

The comprehensive results for each record are listed in Table 2. This table not only details the lock-in function parameters— $\rho_{0.5}$, and ρ_w —but also encompasses essential sedimentological aspects of each record. These aspects include the nature of the sediment (freshwater or marine), the water depth at which the sediment was collected, and the derived accumulation rates from the newly generated age-depth models. Additionally, we include some lithostratigraphic information. It's important to note that this summary provides a small snapshot of the data; it extracts only a fraction of the information available in the original publications. For a more in-depth understanding and comprehensive data, readers are encouraged to refer to the original publications associated with each record.

3.3. Importance of Data Preprocessing

In this section we emphasize the importance of proper data preprocessing and data quality verification. We demonstrate the importance using the BIR record as an example. Panel one and two in (b) of Figure 8 display declination and inclination of BIR after applying our standard preprocessing procedure (see Section 2.4). The estimated lock-in functions after applying our method to the BIR data set are presented in panel (a) of the same figure. Initially, we set an upper bound of 100 cm. Since the parameter values b_4 were close to this upper bound we decided to increase it to 200 cm where we faced the same problem. After increasing it two more times to 400 cm the parameters b_4 were significantly smaller, indicating a successful estimation. The resulting 40 estimated lock-in functions are visualized in (a) of Figure 8. However, our optimizer found an optimal lock-in function with a half lock-in depth of 262.23 cm, corresponding to an offset of $2,745 \pm 339$ years (mean $\pm$ standard deviation with

Table 2

Results Overview and Some Lithostratigraphical and Sedimentological Information Extracted From the Corresponding Publications

Core	$\rho_{0.5}$ (cm)	ρ_w (cm)	Type	Water depth (m)	$\mu_{acc} \pm \sigma_{acc}^a$ (cm/year)	Material ^b
BA3	12.26	15.43	Marine	1,431	0.093 ± 0.036	Upper part: pervasively bioturbated sediment with a fine-grained texture; lower part: fine-grained, diatom-bearing sediment with coarse bedding
BIR	2.73	1.14	Lacustrine	1.5	0.107 ± 0.049	Clay, silty clay
BYE	66.84	105.31	Lacustrine	10.6	0.052 ± 0.008	Partly laminated fine detritus gyttja
ByaP2	59.23	74.58	Lacustrine	10.6	0.052 ± 0.008	Partly laminated fine detritus gyttja
ByaP3	69.9	119.76	Lacustrine	10.6	0.052 ± 0.008	Partly laminated fine detritus gyttja
EIL	1.75	2.64	Lacustrine	6	0.182 ± 0.032	Gray to blackish, layered sediments with moderate TIC and TOC contents
FUR	12.23	41.7	Lacustrine	14.2	0.045 ± 0.011	Partially laminated fine detritus gyttja/clay gyttja, silt
P2	16.69	56.98	Lacustrine	14.2	0.045 ± 0.011	Partially laminated fine detritus gyttja/clay gyttja, silt
P3	0.05	0.1	Lacustrine	14.2	0.045 ± 0.011	Partially laminated fine detritus gyttja/clay gyttja, silt
GYL	38.34	130.9	Lacustrine	17	0.152 ± 0.068	Laminated brown fine detritus gyttja
GP1	53.45	106.9	Lacustrine	17	0.152 ± 0.068	Laminated brown fine detritus gyttja
GP2	28.99	29.95	Lacustrine	17	0.152 ± 0.068	Laminated brown fine detritus gyttja
GP4	32.68	68.01	Lacustrine	17	0.152 ± 0.068	Laminated brown fine detritus gyttja
MOR	0.15	0.45	Lacustrine	–	0.036 ± 0.011	Clay, silty clay
MST	8.45	28.87	Marine	267	0.107 ± 0.049	–
U1305	11.17	27.79	Marine	3,459	0.062 ± 0.016	Ranges from silty clay with sand and little biogenic material, to homogenous nannofossil ooze with silty clay and common bioturbation

Note. For more detailed information see the original publications. ^aMean and standard deviation of accumulation rate derived from our age-depth models. ^bInformation is taken from publications and shortly summarized.

respect to the age depth model). These results are unexpected, as visual inspection of the data suggests a maximum offset of 50 cm.

To investigate this discrepancy, we tried various approaches, including increasing uncertainties, eliminating more data points as outliers, and testing different age-depth models. Nonetheless, these attempts were leading to similar results. After using the E/I-analysis tool on [Palaeomagnetism.org](https://www.palaeomagnetism.org) 2.4.0 (Jollyfant & Pastor-Galán, 2022; M. R. Koymans et al., 2016; M. Koymans et al., 2020) to remove possible inclination shallowing we obtain meaningful results that agree with visual interpretation and expectation. The E/I-analysis yielded a flattening factor of

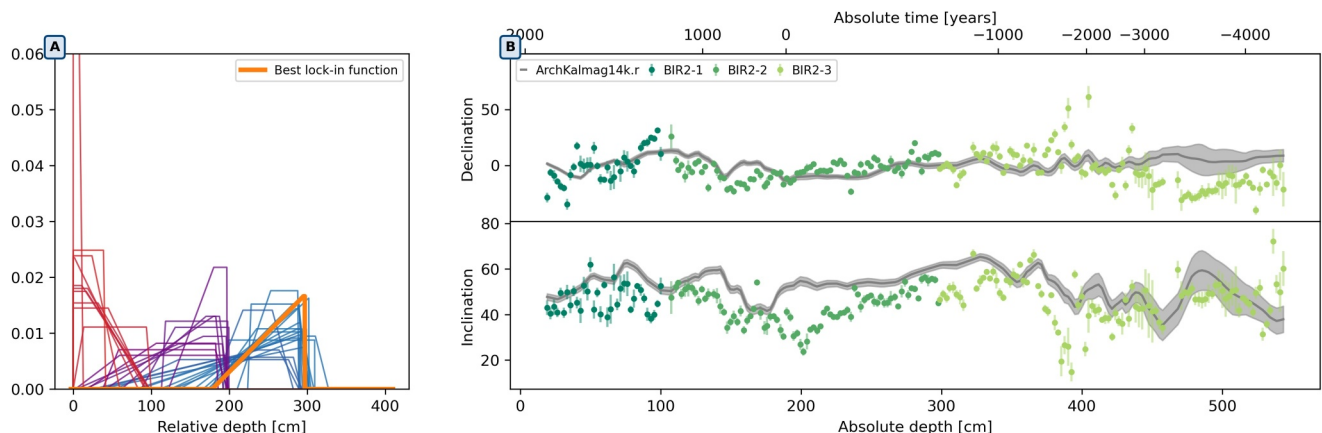


Figure 8. Results for BIR before accounting for inclination shallowing.

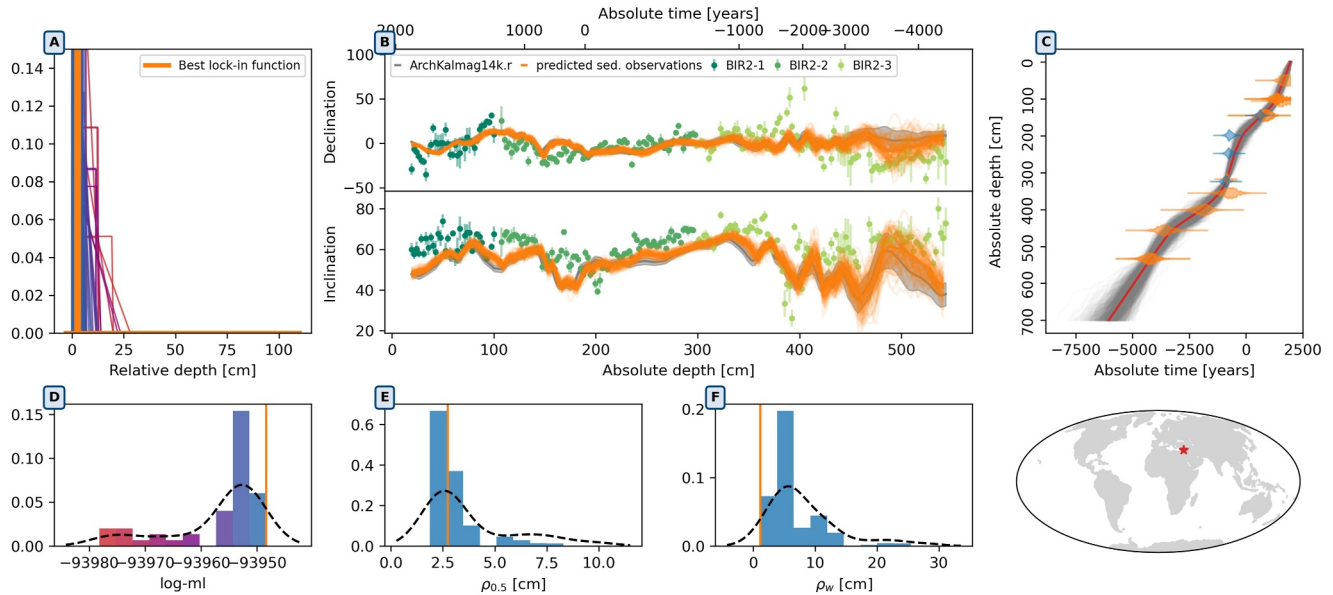


Figure 9. Results for BIR after accounting for inclination shallowing.

$f = 0.54$, indicating significant inclination shallowing (see Figure S12 in Supporting Information S1). To remove inclination shallowing we used the following formula proposed by R. King (1955).

$$\tan(I_o) = f \tan(I_f) \quad (5)$$

where I_o is the observed inclination and I_f the inclination of the external field.

Subsequently, we present the results after applying the flattening factor in Figure 9. The analysis highlights the crucial role of diligent data preprocessing and the importance of considering various factors that can impact the final outcomes of our method.

Removed data points: The first 7 data points from the upper part of the record have been removed due to an abrupt decrease in the inclinations. Additionally, three data points at depth 460, 462, and 464 cm were removed as obvious outlier.

Declination offset: The paleomagnetic data is divided into three sub-sections. The estimated declination offset values are $9.64^\circ \pm 1.36$ for the lower part, $-2.17^\circ \pm 2.3$ for the middle, and $1.53^\circ \pm 1$ for the upper part.

Age-depth model: Radiocarbon dates above 700 cm were taken from the original publication (Schwab et al., 2004) and used to construct the age-depth model. The original publication suggests applying a local reservoir correction of 500–700 years. We decided to employ a correction of 600 years and include an additional error of 200 years to compensate for uncertainty in the reservoir effect estimation. For two depths, both plant remains (orange in Figure 9) and bulk samples (blue) were taken, allowing to estimate an additional reservoir effect of 500 years for the bulk sediment. See the original publication for a discussion of possible mechanisms causing the local reservoir effects.

Estimation: After accounting for inclination shallowing an upper bound of 100 cm turned out to be appropriate. Furthermore, the predicted sediment observations demonstrated a decent correspondence with the actual data, indicating a good fit.

4. Discussion

Applying the proposed method to nine sediment records yields insightful results, demonstrating the general occurrence of offsets and smoothed signals induced by the pDRM process. The range of examples spans from

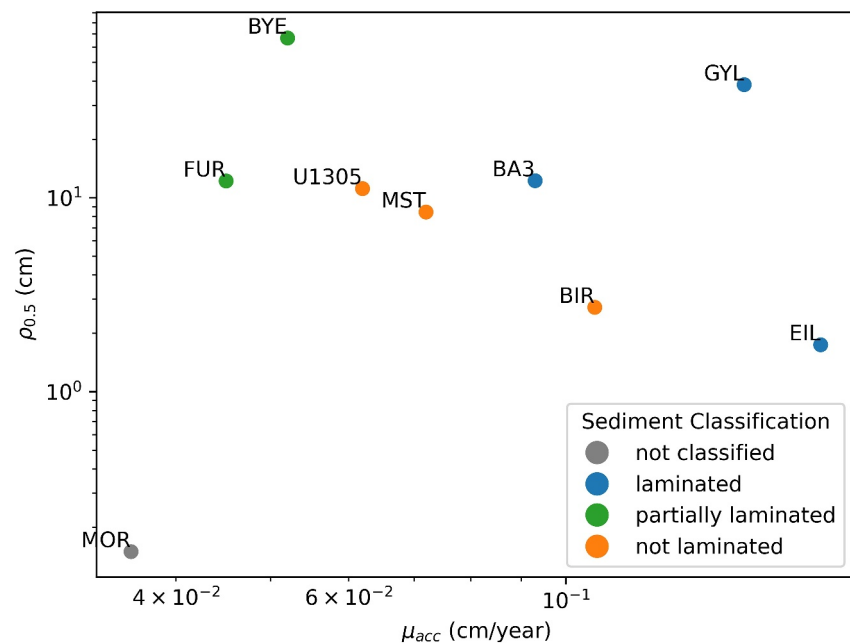


Figure 10. Bilogarithmic plot of mean accumulation rate μ_{acc} versus half lock-in depth $\rho_{0.5}$ colored by sediment classification.

records with almost no distortion (EIL, BIR, MOR) to those with offsets of nearly 70 cm and significant smoothing (see BYE and its sub-cores). While the four parameters b_1 to b_4 are not as precisely determined as the half lock-in depth, the last parameter b_4 , representing the maximal lock-in depth, remains an intriguing parameter to consider. Estimated results of this parameter range from maximum lock-in depths of only a few centimeters to over 130 cm.

While acknowledging that nine examples are insufficient for comprehensive statistical analysis, the observed pDRM effects in six out of the nine sediment records highlight the significance of accounting for this phenomenon. Most of the half lock-in depths of the six records displaying pDRM effects range from 10 to 20 cm. Notably, examples like GYL and BYE demonstrate the potential occurrence of much higher half lock-in depths, reaching 38 and 67 cm. However, it appears that half lock-in depths exceeding 30 cm are less common.

The influence of the number of surrounding archeomagnetic data on the method, was found to be moderate in synthetic tests (Bohsung et al., 2023). In other words, the estimation works even in areas where archeomagnetic and volcanic data is sparse. Yet, as exemplified by MOR, it does affect the uncertainties in predicted sediment observations. This is not surprising and related to the fact that the uncertainties increase with decreasing number of data. This phenomenon is visible across all records as uncertainties are increasing with increasing depth. Several factors are responsible for this. The number of archeomagnetic data constraining the model decreases, the dating of the archeomagnetic records becomes less precise and the age-depth models of sediment records become less reliable.

Investigating potential distinctions between marine and lacustrine sediment records is an interesting question. However, with only three marine records in our data set, a more extensive collection is necessary for a conclusive analysis. Similarly, when it comes to correlating lock-in function parameters with sedimentological or lithostratigraphic features, the current data set remains insufficient for comprehensive examination.

Despite the limited number of data sets available, we have proceeded with preliminary investigations. In Figure 10, we explore the relationship between the mean accumulation rate, denoted as μ_{acc} , and the half lock-in depth, $\rho_{0.5}$. Individual records are distinguished by their classification into “laminated,” “partially laminated,” “not laminated” or “not classified.” Our findings do not indicate any clear differences between laminated and non-laminated sediments, nor do they show a consistent relationship between the half lock-in depth and the accumulation rate, aside from a potential minor decreasing trend. These observations suggest that the mechanisms

influencing DRM and pDRM acquisition are not straightforwardly linked to simple characterizations of sediment lithology or depositional environments. However, it is crucial to note that a more robust data set is essential to substantiate these observations.

Looking ahead, we plan a more detailed and methodical study, aimed at thoroughly exploring these relationships. Our primary focus in this study, though, was to concentrate on the practical application of the newly introduced method across a carefully selected list of sediment records. This initial focus provided a foundation for future, more extensive inquiries. To advance our understanding in these areas, collaboration with experts in sedimentology will be pivotal. Their insights will be instrumental in unraveling the complex interactions and variations inherent in different sedimentary environments.

Notably, the necessity for caution when dealing with sediment records composed of multiple sub-cores. Discrepancies between combined records and sub-cores, exemplified by BYE, FUR, and GYL, emphasize the importance of treating them individually. This recommendation becomes particularly pertinent when significant distances between individual cores may lead to slight variations in sedimentation processes. The individual differences in these cases are discussed in the respective sediment record sections.

Our modeling technique and the global archeomagnetic data set allows the estimation of lock-in function parameters for sediment records from any global location without the need of a pretrained (local) geomagnetic field model. However, our approach relies on the amount, density, and integrity of data fed into the method. In other words, the quality and consistency of the input data act as the bedrock of our modeling technique; any shortfall in the data set, be it sparsity or inconsistency, could impair the accuracy of our results. Likewise, the reliability of age-depth models plays a critical role. We have to rely on the available information, and it is clear that the efficacy of our method depends on the veracity of the archeomagnetic reference data and age information associated to the sediment data. In a future study we will extend the estimation period to more ancient times, where archeomagnetic data and lava flows become increasingly sparse. By extending the estimation period back into such epochs, we will not only assess the resilience of our approach but also explore its potential limits.

One pivotal assumption of our method is the homogeneity of the lock-in function throughout the period under examination. This assumption facilitates a straightforward estimation process and is generally reasonable for short periods and sediment records with minimal lithological variability. However, for long sediment records or those with pronounced lithological differences, this assumption of a homogenous lock-in function may not hold. In such cases, it could be beneficial to calculate distinct lock-in functions for individual segments of the sediment core. This segment-specific approach allows for a more nuanced capture of the varying pDRM effects that may occur due to changes in sedimentation processes or material properties. When segmenting the sediment record, however, it is essential to consider the archeomagnetic data's spatial distribution. Since archeomagnetic data points are not uniformly spread across the globe, and their density tends to diminish with increasing age, there may be limitations in the data available to inform the lock-in functions for each segment. Consequently, any segmentation strategy must be balanced with the availability and quality of archeomagnetic data to ensure the reliability of the estimated lock-in functions.

5. Conclusion

In this paper, we applied the method presented in Bohsung et al. (2023) to analyze nine sediment records from different locations worldwide. The results reveal the presence of distortions associated with the lock-in or pDRM process in six out of the nine investigated sediment records.

In addition to the investigation of the pDRM process, we propose a new method for estimating the offset required to transferring relative to absolute declinations. Our method uses the ArchKalmag14k.r model (Schanner et al., 2022), which relies solely on archeomagnetic data. The proposed Bayesian modeling technique is able to take uncertainties into account, making the estimated offset more reliable.

The example of BIR underscores the significance of meticulous data preprocessing and the consideration of distortions beyond pDRM that may impact the data. This includes an accurate estimation of absolute declination and the consideration of inclination shallowing.

Motivated by our findings, we will work on involving simultaneous estimation of declination offset parameters and the shallowing factor, along with lock-in function parameters. In other words, we will estimate all parameters

related to these effects simultaneously, resulting in a comprehensive sediment data preprocessing software for general applicability.

Furthermore, we will explore the deconvolution and application of the estimated parameters to the sediment data. Our goal is to develop a specialized sediment preprocessing software, enhancing the reliability of sediment data for geomagnetic field modeling.

Data Availability Statement

All data (except for MST) used in this study as well as a python implementation of the method can be found in the GitLab repository (Bohsung & Schanner, 2023). The data for the MST record is not publicly available. The author send us the data but did not agree to publish the raw data. Therefore, we published our results but you will not find it in the repository. To reproduce the results you have to ask the author for the data and then use our preprocessing routine. On our website (<https://sec23.git-pages.gfz-potsdam.de/korte/pdrm/>) jupyter notebooks have been published that can be used to reconstruct the results presented in this paper or to apply the method to additional data. The raw data for records from GEOMAGIA can be found on GEOMAGIA (Brown et al., 2015). More results can be found in Supporting Information S1.

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